

MINUSINSK, Russian Federation

WMO: 298660

Lat: 53.717N

Lon: 91.700E

Elev: 835

StdP: 14.26

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-31.9	-40.0	0.6	-35.9	-35.6	0.7	-31.8	18.8	20.8	16.9	19.4	0.0	110	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	87.7	66.6	84.5	65.1	81.3	64.1	68.9	82.7	67.4	79.8	65.9	77.2	5.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	94.1	72.8	63.3	89.6	71.9	61.7	84.9	71.2	33.7	82.9	32.4	79.7	31.2	77.2	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.8	13.6	DB	-36.9	93.6	6.1	3.2	-41.3	95.9	-44.8	97.7	-48.2	99.5	-52.6	101.9	(4)
(5)			WB	-37.0	72.4	6.0	2.5	-41.3	74.2	-44.8	75.7	-48.2	77.1	-52.6	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	-3.7	2.7	24.1	42.1	50.5	64.6	67.3	63.6	50.6	37.4	20.1	6.1		
	DBStd	26.43	15.69	15.11	12.99	8.58	8.18	6.72	4.86	5.61	7.18	8.65	13.38	14.96		
	HDD50	6888	1666	1325	804	261	99	1	0	0	77	398	896	1361		
	HDD65	10960	2131	1745	1268	688	453	87	31	95	434	856	1346	1826		
	CDD50	1637	0	0	0	23	114	439	538	421	95	7	0	0		
	CDD65	234	0	0	0	0	3	75	103	51	2	0	0	0		
	CDH74	3133	0	0	0	12	161	1130	1101	643	86	0	0	0		
	CDH80	954	0	0	0	1	34	380	331	183	25	0	0	0		
(14) Wind	WSAvg	4.7	3.0	3.6	5.1	6.0	6.4	4.8	4.3	4.0	4.4	5.2	5.5	4.6		
(15) Precipitation	PrecAvg	14.0	0.3	0.3	0.2	0.6	1.4	2.2	2.6	2.5	1.8	1.0	0.6	0.5		
	PrecMax	20.9	1.3	1.2	1.0	1.5	3.7	7.1	5.4	7.2	4.7	2.4	1.9	1.2		
	PrecMin	8.7	0.0	0.0	0.0	0.1	0.1	0.5	0.8	0.7	0.2	0.0	0.0	0.1		
	PrecStd	2.4	0.2	0.2	0.2	0.4	0.7	1.1	1.1	1.4	0.9	0.5	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	38.4	58.4	75.6	84.2	92.1	91.5	90.1	83.4	66.5	50.0	37.1		
		MCWB	29.9	33.1	43.4	53.3	60.2	67.3	69.6	68.3	62.1	52.4	42.3	30.7		
	2%	DB	30.4	34.7	51.5	69.5	78.4	87.8	87.1	84.9	74.8	60.1	45.2	33.5		
		MCWB	26.9	30.7	40.4	51.5	56.8	65.6	67.1	67.2	60.4	50.0	38.8	29.4		
	5%	DB	26.1	31.3	46.4	64.3	73.5	84.6	83.5	81.1	69.5	55.6	40.9	30.2		
		MCWB	23.1	28.1	38.0	48.8	54.7	64.1	66.0	65.5	57.4	47.0	35.8	26.8		
	10%	DB	20.1	26.2	41.9	59.2	68.0	80.9	80.2	77.0	64.8	51.2	37.1	27.3		
		MCWB	17.6	23.2	34.8	46.2	52.8	63.2	64.9	63.6	55.1	43.8	33.0	24.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.2	34.0	44.2	55.4	62.1	70.5	71.4	70.7	64.7	53.7	43.3	33.0		
		MCDB	33.3	37.6	56.1	74.1	79.5	85.9	84.3	84.5	80.1	63.3	48.9	36.5		
	2%	WB	27.0	31.2	41.0	52.2	59.1	68.2	69.0	68.3	61.4	50.6	39.4	29.2		
		MCDB	30.3	34.1	50.5	66.9	74.2	82.4	82.7	80.7	72.0	59.1	44.1	32.9		
	5%	WB	23.3	27.8	38.3	49.8	56.7	66.4	67.5	66.4	59.0	47.6	36.1	26.9		
		MCDB	26.1	31.5	46.1	62.4	69.9	79.6	79.5	77.5	68.4	54.6	40.3	30.3		
	10%	WB	17.7	22.8	35.2	47.0	54.1	64.5	66.2	64.9	56.2	44.6	33.2	24.4		
		MCDB	19.9	26.1	41.5	58.3	66.2	77.1	77.1	74.9	62.8	50.3	37.2	27.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	23.8	22.8	23.8	23.9	24.7	20.8	22.0	22.8	18.7	14.7	16.5		
		MCDBR	18.3	21.0	27.1	34.9	36.1	33.3	29.3	30.0	33.3	28.4	15.9	14.4		
	5% WB	MCWBR	16.0	17.0	16.9	18.0	17.0	13.2	11.8	13.2	18.6	17.8	10.9	12.0		
		MCWBR	18.1	20.3	25.9	32.4	29.7	27.3	24.2	26.0	29.8	28.2	15.4	13.9		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.319	0.368	0.409	0.416	0.405	0.417	0.403	0.365	0.348	0.323	0.289		
	taud		2.052	2.005	1.977	2.134	2.176	2.311	2.299	2.309	2.351	2.269	2.111	2.002		
	Ebn at Noon		204	235	250	258	264	268	262	259	254	228	186	170		
	Edh at Noon		23	34	44	43	43	38	38	36	30	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		319	647	1110	1416	1676	1929	1746	1461	1026	569	284	207		
	RadStd		24	45	88	79	162	167	115	114	97	51	32	21		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page